NEXUS-1 — Phase 0

Nuclear Plant Operator Console & Digital-Twin Platform

Functional Guide - Phase 0 of the project. An analytical walk-through of what the application does, how each screen works, and how the parts connect into a single behaving system. This guide documents **Phase 0**, the demonstration build, and is intended to accompany it. The phased plan toward a production system is in the separate Project Roadmap.

> **SCOPE - READ FIRST**
>
> NEXUS-1 is a **demonstration and educational project**. It is not connected to any real reactor, sensor, or control system, and its models are studied and representative rather than validated or certified. Throughout the application and this guide, anything labelled illustrative is exactly that. The strength of the project is in its software design, simulation, visualisation, and systems thinking - not in claiming nuclear-engineering validity.

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1. Introduction and operating context

NEXUS-1 is an interactive operator console and digital-twin platform for a nuclear power facility, delivered as a single self-contained web application. It presents a complete plant - reactor physics, power generation, the secondary steam and electrical systems, safety and radiation monitoring, a multi-unit fleet, component ageing and decommissioning, staffing, a robotic fleet, zone access control, and analytical tools - through one navigable interface driven by a live in-browser simulation.

The design follows the operating philosophy of real digital-twin systems. The platform is conceived as a monitoring, analytics, optimisation and decision-support layer that observes and advises but holds **no control authority** over certified safety systems. Several operating modes frame this intent: **Simulation** (a fully virtual plant, which is what this build is), **Training**, **Shadow** (mirroring live data without acting), **Advisory** (recommendations to humans), **Supervisory** (integrated monitoring under human supervision), and **Hybrid Digital Twin** (a simulation kept in step with live data).

2. Getting started

Running it. Open the application's HTML file in any modern web browser (Chrome, Edge, Firefox or Safari). There is no installation, account or server - the entire application runs locally in the browser.

Internet note. One screen, the 3D Reactor View, uses a 3D graphics library loaded from a public content-delivery network, so it needs an internet connection to render. If the library cannot load, that single screen shows a short fallback message and everything else continues to work offline.

> **TIP**
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> A good first tour: open **Reactor > Control Rods** and move the rod bank, then press **SCRAM**. Now visit **Incident Analysis** and **Compliance** to see the event recorded, and **Rod Inspection** to see the rods flagged for post-trip inspection. This single sequence demonstrates how the screens are connected.

3. The interface: layout, navigation and conventions

Layout. A fixed left **sidebar** lists every screen, grouped by area; expandable groups (Reactor, Rod Inspection, Plant Lifecycle, Robotics & Vehicles, Zone Access) open to reveal their sub-screens. A **top bar** shows the current screen title, a **unit selector** for choosing which reactor unit you are viewing, a plant-state indicator, a clock, and a **'?' help button**. The main area shows the selected screen.

Navigation. Click any sidebar item to open its screen. Click a group header to expand or collapse it. The unit selector in the top bar changes which unit the unit-specific screens display.

Conventions. Status colour is consistent everywhere: **green** = nominal, **amber** = degrading or caution, **red** = a limit reached or the reactor tripped. Numeric fields update live as the simulation runs.

Getting help in context. The **'?'** button (top right) opens a small panel describing the screen you are currently viewing and its key points. The sidebar's **Help & Guide** screen gives the full in-application overview, and **About** states the author and scope.

4. Whole-plant and fleet views

Plant Overview

What it does. A single-screen snapshot of the selected unit's headline parameters and overall health. **Using it.** Switch units with the top-bar selector to see each unit's state. **Connections.** Reflects the selected unit; the plant-state indicator updates immediately on selection.

Plant Fleet

What it does. Presents a multi-unit, modular fleet - two large pressurised-water reactors (PWR) and three small modular reactor (SMR) modules - and aggregates their output onto a shared grid. **Using it.** Toggle individual units on or off; pick the active unit. Each unit runs its own simulation scaled to its capacity. **Connections.** The active unit chosen here (or via the top-bar selector) is the unit that the Core, Control Rods, Aging, Decommissioning and 3D screens display. Aggregate output and grid frequency respond as units come on and off.

5. The reactor: core, rods, neutronics, kinetics and the 3D twin

Reactor Core

What it does. Shows the core layout as a map of fuel assemblies and the overall core state. **Key behaviour.** The map redraws for the selected unit - a full core for PWR units and a visibly smaller core for SMR modules - and its colours follow the unit's live power, cooling toward blue when a unit is offline. **Connections.** Driven by the selected unit and its power level.

Control Rods

What it does. The interactive heart of the operational simulation. Moving the rod bank changes reactivity, which drives neutron flux, thermal power and electrical output; a **SCRAM** button performs an emergency shutdown. **Using it.** Drag the rod slider to raise or lower power; press SCRAM to trip the unit. **Connections.** A SCRAM here propagates across the application - it is written to the Incident Analysis timeline and the Compliance log, counted as a safety-rod demand cycle in Rod Inspection, and added as a thermal-fatigue cycle in the unit's Aging page.

Neutronics

What it does. Detail on neutron flux distribution and the components that make up reactivity for the selected unit.

Reactor Kinetics

What it does. A physically faithful point-kinetics simulator: six delayed-neutron precursor groups with Doppler, moderator and xenon feedback, and a coupled fuel/coolant thermal model, integrated with a numerically stable scheme. **Using it.** Apply small reactivity steps and watch the prompt jump followed by feedback self-regulation; press SCRAM to see power fall quickly but not instantly (the delayed-neutron tail); switch the time scale up to 600x after a shutdown to watch the xenon transient build over hours. **Why it matters.** It reproduces the genuine, non-obvious phenomena of reactor dynamics rather than a scripted animation.

> **SCOPE**
>
> The Reactor Kinetics model uses the real equation structure and standard delayed-neutron data, but representative (textbook-order) constants rather than values tuned to a specific reactor. It is an educational simulator, not a licensed analysis code.

3D Reactor View

What it does. Two three-dimensional cores side by side, presenting the selected unit as a digital twin: the left core is the unit seen through (simulated) sensors - noisy and slightly lagged - and the right core is its clean expected model. **Using it.** The controls drive the selected unit's rods (insertion, withdraw, SCRAM); drag a core to rotate it; fuel colour indicates temperature. **Connections.** Both cores rebuild to the selected unit's size (smaller for SMR modules), and a SCRAM here is recorded like any other. The **Twin Divergence** panel quantifies the gap between measured and modelled - the early-warning signal a real twin provides.

6. Secondary side and electrical output

Steam & Secondary

What it does. The secondary circuit - steam generator levels and pressure, and the path toward the turbine.

Coolant

What it does. The primary coolant loops: flows and temperatures (hot and cold legs). These quantities feed the System Dependencies model.

Power & Grid

What it does. Electrical generation and grid synchronisation - generator output, frequency, and related electrical quantities, aggregated across online units.

7. Monitoring, events and safety records

Radiation

What it does. Radiation and dose monitoring against limits.

Alarms & Events

What it does. Active alarm management - acknowledging and tracking alarm state across the plant.

Incident Analysis

What it does. A timeline of events. Live SCRAM events are inserted automatically at the top as they occur, above a baseline event reconstruction. **Connections.** Fed by the shared event layer whenever a SCRAM is triggered anywhere in the application.

Compliance / Audit

What it does. A tamper-evident, hash-sealed record of operator and system actions. **Connections.** Each SCRAM writes a safety-type entry and a system entry here automatically, giving the regulatory-style trail a real cause.

OT Security

What it does. The operational-technology security posture - zero-trust, network segmentation, hardware security modules, multi-factor authentication and intrusion detection status - reflecting the critical-infrastructure protection a real deployment requires.

8. Analysis and intelligence

AI Diagnostics

What it does. Supervised predictive diagnostics - failure-probability and degradation forecasts presented as advice. It answers 'what is the probability of failure?' and is distinct from the reinforcement-learning optimiser below.

Optimization (RL) - experimental

What it does. An adjacent, clearly-labelled **BETA, twin-only, advisory** capability. A reinforcement-learning agent learns a rod-control policy by trial and error inside the simulation, to hold a target power with minimal control-rod movement. **Using it.** Press **Train policy** and watch the reward curve climb; the learned policy is displayed as a readable table (rows = power deviation, columns = trend), so it is interpretable rather than a black box. The **Advisory** panel then recommends an action for the current state, with a plain-language explanation and a safety clamp; it never takes control of a real plant.

> **SCOPE**
> The RL agent is a genuine tabular Q-learning learner trained against a fast reduced-order surrogate of the reactor model. It is deliberately interpretable, advisory-only, and trains exclusively in simulation - mirroring real industry practice of training control policies only in a twin.

System Dependencies

What it does. Makes the plant's interconnections explicit, in three tabs. The **Dependency Graph** is a live, colour-coded network of components; click any node to trace everything it influences downstream, with approximate propagation delays, and feedback loops drawn distinctly. The **Influence Matrix** is an engineer's table of influence weights; click a cell to read how strongly one component affects another. The **Causal Chain** reconstructs a reactor-trip sequence step by step and offers a 'what if the valve were repaired?' counterfactual. **Connections.** The graph's node colours are driven by the live reactor state.

Digital Twin & Trends

Digital Twin reconciles model expectation against (simulated) telemetry. **Trends** shows historical time-series of key parameters; like other charts, it follows the selected unit.

9. Inspection and material integrity (NDT)

Rod Inspection

What it does. Non-destructive testing for each rod type, including a procedurally rendered radiograph (X-ray) viewer and the six standard NDT methods. The application correctly represents real engineering constraints - for example, magnetic particle testing is shown as not applicable on the non-ferromagnetic fuel cladding. **Connections.** For the safety and control rods, the screen shows the number of SCRAM demand cycles accrued this session and flags them for post-trip inspection - a direct link from operational events to inspection workload.

10. Plant lifecycle: ageing and decommissioning

Aging & Degradation

What it does. For the selected unit, tracks the major long-lived components (reactor vessel, steam generators, primary piping, containment concrete, cables, core internals) - each with its degradation mechanism, percentage of design life consumed, a trend, a remaining-life estimate, and an OK/Monitor/Limiting status. **Connections.** The effective age grows with simulated operating time, and each SCRAM adds thermal-fatigue cycles that nudge the relevant components' life-consumed upward. Older units show real wear; new SMR modules read pristine.

Decommissioning

What it does. The end-of-life lifecycle. A fleet table lets you begin decommissioning a unit, which shuts it down and moves it through stages (final shutdown, defueling, decontamination, safe enclosure, dismantling, waste treatment, site release), with the radiological inventory decaying over time and waste volumes accumulating by category.

11. People, robots and zone access

Personnel & Staffing Stress Test

What it does. Sector rosters with critical-role coverage, and a stress test that applies absence scenarios (illness, night shift, targeted gaps, pandemic) and reports whether the minimum shift complement is still MET, DEGRADED or BREACHED.

Robotics & Vehicles + Mission Readiness

What it does. A robotic and vehicle fleet organised by category, each asset tracked for operational status, battery charge and cumulative radiation dose (robot electronics are dose-limited). The Mission Readiness test applies scenarios and reports whether each standard mission can still be covered by a capable, available, charged and dose-eligible asset.

Zone Access

What it does. Tracks people, vehicles and robots together, each with a simulated access tag. **Live Presence** shows who and what is in each zone over a defence-in-depth diagram, flags anyone present in a zone they are not cleared for, and can simulate movement. The **Permissions Matrix** documents which class may enter which zone.

12. How NEXUS-1 behaves as a connected system

A central design goal is that the screens are not independent dashboards but a single behaving system. The most important couplings are:

- **A SCRAM has consequences.** Triggering a SCRAM (from Control Rods or the 3D view) writes a live entry to Incident Analysis, two hash-sealed entries to Compliance, a demand-cycle counter and inspection flag in Rod Inspection, and a thermal-fatigue cycle in that unit's Aging page.
- **Operation accrues life.** While a unit is online, simulated operating time accumulates and feeds the unit's effective age in the Aging model.
- **The unit selector propagates.** Choosing a unit updates the Overview, Core (size and colour), Control Rods, gauges and charts, the Fleet highlight, the plant-state indicator, the Aging and Decommissioning detail, and the 3D Reactor View.
- **Live state drives analysis.** The System Dependencies graph colours its nodes from the live reactor state, so operational actions visibly ripple through the dependency map.

By design, a separate single-core physics engine powers **Reactor Kinetics, Optimization (RL) and the System Dependencies status**, independently of the fleet unit selector; the 3D Reactor View additionally mirrors the selected unit's size and state for presentation. Screens that are plant-wide by nature - Alarms, Incident Analysis, Compliance, OT Security, Personnel, Robotics, Zone Access - are intentionally not unit-specific.

13. Architecture and implementation

The application is a single self-contained HTML document with CSS and JavaScript and no external framework or build step. All diagrams, gauges, charts, the radiograph and the reactor cross-sections are hand-written SVG; the three-dimensional cores use a WebGL library loaded at runtime. The simulation is organised as cooperating modules sharing a small event layer, so that actions in one area (such as a SCRAM) can be observed by the others. The Reactor Kinetics screen implements the point-kinetics equations directly; the optimisation screen implements a genuine tabular Q-learning agent; the dependency, lifecycle, staffing, robotics and access screens are rule-based models. The intended production architecture - described in the roadmap - is a set of .NET microservices with real-time streaming, an Angular front end, a message bus and a time-series store.

14. What is real versus simulated

Real / genuine	Simulated / illustrative
The interface, navigation, visualisations and interactions are real, working code.	All data is generated in the browser; nothing connects to a real plant, sensor or control system.
The reactor-kinetics physics and the reinforcement-learning agent are real in structure and genuinely compute/learn.	Their constants are representative, not tuned to a specific reactor; the agent trains on a reduced-order surrogate.
The workflows reflect real industry concepts (NDT methods, staffing rules, dose limits, decommissioning stages, defence in depth).	Ageing, dependency weights, radiological figures, names, ratings and dates are plausible placeholders.

15. Roadmap to a real system

- Move the simulation server-side (.NET) and stream it to the interface in real time.
- Persist telemetry to a time-series store and serve real history and trends.
- Introduce a message bus and split the system into services (telemetry, twin, alarms, audit).
- Add industrial-protocol gateways (OPC-UA first), strictly read-only from the plant; keep analytics and AI advisory only.
- Harden for the domain: redundancy and high availability, operational-technology security, and the formal verification and certification a real deployment requires.

16. About, scope and licence

NEXUS-1 was created by **Grigorios Agathangelidis**, whose background is in Electrical and Software Engineering, as a personal project: an effort to immerse in the nuclear-energy domain, study it thoroughly, and apply software-architecture, simulation, visualisation, user-interface and systems-design skills to it. It does not claim nuclear-engineering expertise, and it is careful to mark where its models are illustrative rather than validated.

 The application is a demonstration and educational tool; it is not a licensed analysis tool and is not connected to any real reactor.